

Molecular Dynamics

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**Molecular dynamics (MD) is an archetype of scientific computing;
use it to learn parallel computing & visualization**

Objectives: Understand

1. md.c line by line
2. simple MD principles well
3. computational complexity & flop/s (assignment 1)



Father of Molecular Dynamics

PHYSICAL REVIEW

VOLUME 136, NUMBER 2A

19 OCTOBER 1964

Correlations in the Motion of Atoms in Liquid Argon*

A. RAHMAN

Argonne National Laboratory, Argonne, Illinois

(Received 6 May 1964)

A system of 864 particles interacting with a Lennard-Jones potential and obeying classical equations of motion has been studied on a digital computer (CDC 3600) to simulate molecular dynamics in liquid argon at 94.4°K and a density of 1.374 g cm⁻³.

Aneesur Rahman—Father of molecular dynamics

Argonne physicist Aneesur Rahman, known worldwide as the “father of molecular dynamics,” pioneered the application of computer science to physical systems.

In 1960, Rahman successfully modeled the behavior of a cluster of 864 argon atoms on a computer that could perform only 150,000 calculations per second. **150 Kflop/s!**

While Argonne's new IBM Blue Gene® /P supercomputer runs nearly 3 million times faster than Rahman's CDC 3600, today's scientists still base the code for their models on Rahman's algorithms.

Since 1993, the [American Physical Society](#) has annually awarded the [Aneesur Rahman Prize](#) for outstanding achievement in computational physics research.

See the Nobel lecture by Michael Levitt [<https://aiichironakano.github.io/phys516/levitt-lecture-slides.pdf>]

See Berkeley CS 267 HW2 [<https://sites.google.com/lbl.gov/cs267-spr2025>]



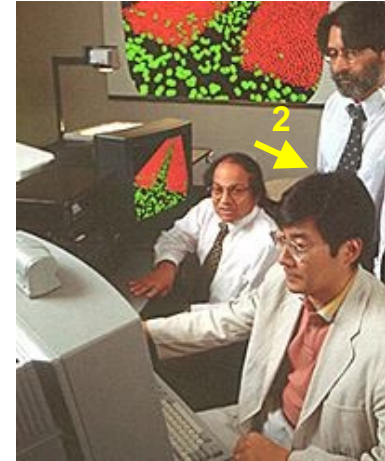
Your Rahman Number?



Anees Rahman ('67)



Anees Rahman & Priya Vashishta
at Argonne National Lab ('81)



Priya Vashishta,
Rajiv Kalia
and AN ('02)

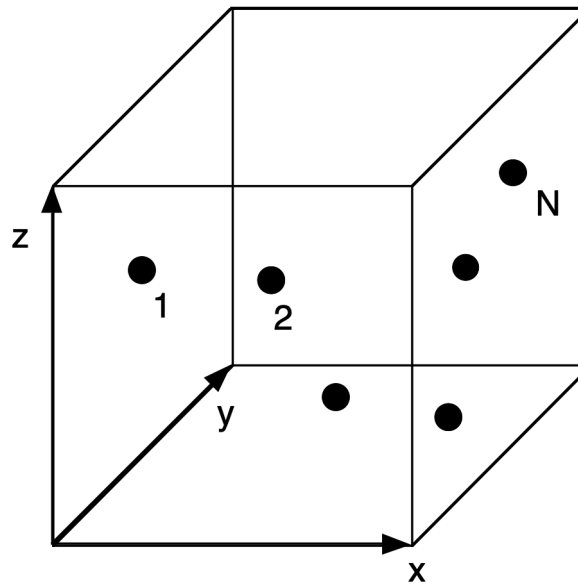
<https://aiichironakano.github.io/cs596/Battimelli-ComputerMeetsPhysics-Springer20.pdf>, pp. 58 & 128

Your Rahman number is 3

System: A Set of Point Atoms

$$\{\vec{r}_i = (x_i, y_i, z_i) \mid x_i, y_i, z_i \in \mathbb{R}, i = 0, \dots, N - 1\}$$

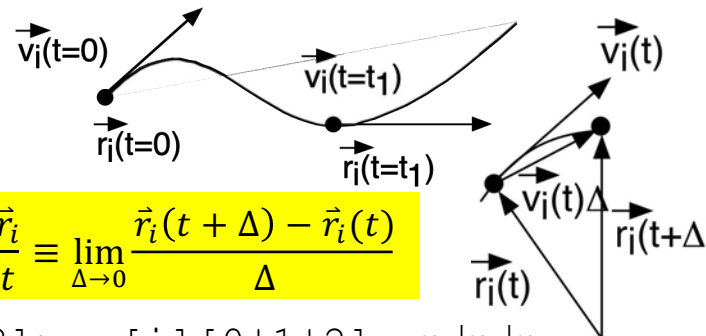
```
md.c  int nAtom: N, # of atoms  
      NMAX: Max # of atoms  
      double r[NMAX][3]: r[i][0|1|2] = x_i|y_i|z_i
```



See lecture on [basic MD algorithm](#) for math symbol vs. C variable correspondence

Trajectory

Trace of atom positions



Velocity

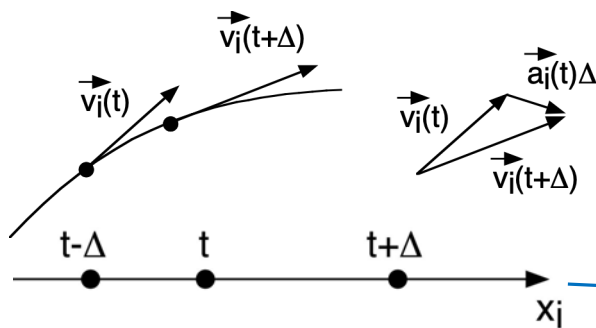
$$\vec{v}_i(t) = \dot{\vec{r}}_i(t) = \frac{d\vec{r}_i}{dt} \equiv \lim_{\Delta \rightarrow 0} \frac{\vec{r}_i(t + \Delta) - \vec{r}_i(t)}{\Delta}$$

```
double rv[NMAX][3]: rv[i][0|1|2] = vix|viy|viz
```

Acceleration

$$\vec{a}_i(t) = \ddot{\vec{r}}_i(t) = \frac{d^2\vec{r}_i}{dt^2} = \frac{d\vec{v}_i}{dt} \equiv \lim_{\Delta \rightarrow 0} \frac{\vec{v}_i(t + \Delta) - \vec{v}_i(t)}{\Delta}$$

```
double ra[NMAX][3]: ra[i][0|1|2] = aix|aiy|aiz
```



$$\begin{aligned} \vec{a}_i &= \lim_{\Delta \rightarrow 0} \frac{\vec{v}_i(t + \Delta/2) - \vec{v}_i(t - \Delta/2)}{\Delta} \\ &= \lim_{\Delta \rightarrow 0} \frac{\frac{\vec{r}_i(t + \Delta) - \vec{r}_i(t)}{\Delta} - \frac{\vec{r}_i(t) - \vec{r}_i(t - \Delta)}{\Delta}}{\Delta} \\ &= \lim_{\Delta \rightarrow 0} \frac{\vec{r}_i(t + \Delta) - 2\vec{r}_i(t) + \vec{r}_i(t - \Delta)}{\Delta^2} \end{aligned}$$

3-point difference

Newton's Equation of Motion

Trajectory is determined by
Newton's 2nd law:

$$m\ddot{\vec{r}}_i(t) = \vec{F}_i(t)$$

Initial value problem: Given initial particle positions & velocities, $\{(\vec{r}_i(0), \vec{v}_i(0))\}$, obtain those at later times $\{(\vec{r}_i(t), \vec{v}_i(t)); t > 0\}$

Potential energy:

$$\vec{F}_i = -\frac{\partial}{\partial \vec{r}_i} V(\vec{r}^N) = -\left(\frac{\partial V}{\partial x_i}, \frac{\partial V}{\partial y_i}, \frac{\partial V}{\partial z_i}\right)$$

where the partial derivative is

$$\frac{\partial V}{\partial x_k} = \lim_{h \rightarrow 0} \frac{V(x_0, y_0, z_0, \dots, \boxed{x_k + h}, y_k, z_k, \dots, x_{N-1}, y_{N-1}, z_{N-1}) - V(x_0, y_0, z_0, \dots, \boxed{x_k}, y_k, z_k, \dots, x_{N-1}, y_{N-1}, z_{N-1})}{\boxed{h}}$$

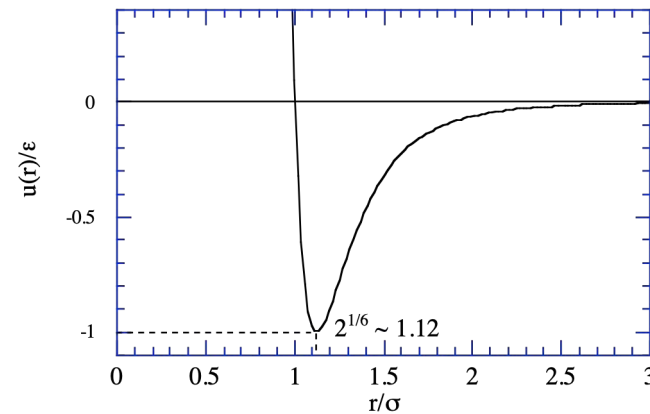
Lennard-Jones Potential

Sum over $\forall$ distinct pairs

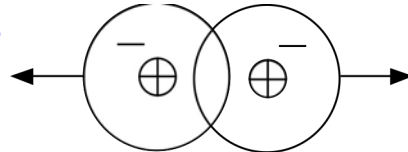
$$V(\vec{r}^N) = \sum_{i < j} u(r_{ij}) = \sum_{i=0}^{N-2} \sum_{j=i+1}^{N-1} u(r_{ij})$$

where $\vec{r}_{ij} = \vec{r}_i - \vec{r}_j$; $r_{ij} = |\vec{r}_{ij}|$ and

$$u(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$



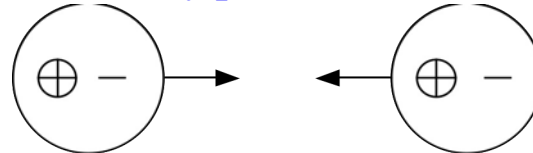
Short-range repulsion by Pauli exclusion between electrons



Long-range attraction by polarization



Johannes D. van der Waals
Nobel Physics Prize (1910)



Wolfgang Pauli
Nobel Physics Prize (1945)



John E. Lennard-Jones
PPS 43, 461 (1931)

Normalization

For Argon atoms:

Atoms are tiny & fast

> $m = 6.6 \times 10^{-23}$ gram

> $\varepsilon = 1.66 \times 10^{-14}$ erg (erg = gram•cm²/second²)

> $\sigma = 3.4 \times 10^{-8}$ cm

Define length, energy & time units as

$$\begin{cases} \vec{r}_i = \vec{r}'_i \sigma = 3.4 \times 10^{-8} [\text{cm}] \times \vec{r}'_i \\ V = V' \varepsilon = 1.66 \times 10^{-14} [\text{erg}] \times V' \\ t = \sigma \sqrt{m/\varepsilon} t' = 2.2 \times 10^{-12} [\text{sec}] \times t' \end{cases}$$

The equation of motion in these units:

$$\begin{aligned} \frac{d^2 \vec{r}_i}{dt^2} &= -\frac{\partial V}{\partial \vec{r}_i} = \vec{a}_i \\ V(\vec{r}^N) &= \sum_{i < j} u(r_{ij}) \\ u(r) &= 4 \left(\frac{1}{r^{12}} - \frac{1}{r^6} \right) \end{aligned}$$

$$\begin{aligned} &1.66 \times 10^{-14} \text{ erg} \\ &= 3.97 \times 10^{-25} \text{ kcal} \\ &= 7.35 \times 10^{-28} \text{ Big Mac} \end{aligned}$$



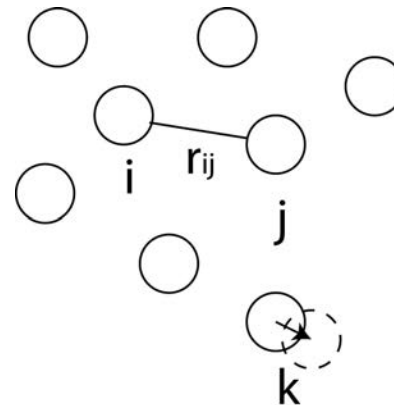
540 kcal

Analytic Force Formula

Chain rule: $\vec{a}_k = -\frac{\partial}{\partial \vec{r}_k} \sum_{i < j} u(r_{ij}) = -\sum_{i < j} \frac{\partial r_{ij}}{\partial \vec{r}_k} \frac{du}{dr_{ij}}$

$$\begin{aligned} \frac{\partial r_{ij}}{\partial \vec{r}_k} &= \left(\frac{\partial}{\partial x_k}, \frac{\partial}{\partial y_k}, \frac{\partial}{\partial z_k} \right) \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2} \\ &= \frac{(2(x_i - x_j), 2(y_i - y_j), 2(z_i - z_j))}{2\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}} (\delta_{ik} - \delta_{jk}) \quad \delta_{ij} = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases} \\ &\quad \left(\because \frac{d}{dx} [f(x)]^{1/2} = \frac{1}{2} [f(x)]^{-1/2} \frac{df}{dx} \right) \end{aligned}$$

$$\begin{aligned} \frac{du}{dr} &= 4 \left(-\frac{12}{r^{13}} + \frac{6}{r^7} \right) = -\frac{48}{r} \left(\frac{1}{r^{12}} - \frac{1}{2r^6} \right) \\ &\quad \left(\because \frac{d}{dr} r^{-n} = -nr^{-n-1} \right) \end{aligned}$$



For chain rule, see Sec. 6.5.2 in “Deep Learning” (<https://www.deeplearningbook.org>)

Molecular Dynamics Problem

Given initial atomic positions & velocities, $\{(\vec{r}_i(0), \vec{v}_i(0)) | i = 0, \dots, N-1\}$,
 obtain those at later times, $\{(\vec{r}_i(t), \vec{v}_i(t)) | i = 0, \dots, N-1; t > 0\}$,
 by integrating the ordinary differential equation,

$$\ddot{\vec{r}}_k(t) = \ddot{\vec{a}}_k(t) = -\frac{\partial}{\partial \vec{r}_k} \sum_{i < j} u(r_{ij}) = \sum_{i < j} \vec{r}_{ij}(t) \left(-\frac{1}{r} \frac{du}{dr} \right)_{r=r_{ij}(t)} (\delta_{ik} - \delta_{jk})$$

where

$$-\frac{1}{r} \frac{du}{dr} = \frac{48}{r^2} \left(\frac{1}{r^{12}} - \frac{1}{2r^6} \right)$$

$$\vec{r}_{ij}(t) = \vec{r}_i(t) - \vec{r}_j(t)$$

$$r_{ij}(t) = |\vec{r}_{ij}(t)|$$

$$\delta_{ik} = \begin{cases} 1 & i = k \\ 0 & i \neq k \end{cases}$$

	0	1	2	3
0				
1				
2				
3				

Force calculation algorithm— $O(N^2)$:

for $k = 0$ to $N-1$, $\ddot{\vec{a}}_k = 0$

for $i = 0$ to $N-2$

for $j = i+1$ to $N-1$

compute $\ddot{\vec{a}} = \vec{r}_{ij} \left(-\frac{1}{r} \frac{du}{dr} \right)_{r=|\vec{r}_{ij}|}$

$\ddot{\vec{a}}_i += \ddot{\vec{a}}$

$\ddot{\vec{a}}_j -= \ddot{\vec{a}}$ Newton's 3rd law

```

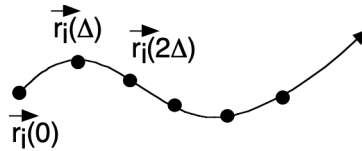
for (n=0; n<nAtom; n++)
  for (k=0; k<3; k++) ra[n][k] = 0.0;
for (j1=0; j1<nAtom-1; j1++) {
  for (j2=j1+1; j2<nAtom; j2++) {
    for (rr=0.0, k=0; k<3; k++) {
      dr[k] = r[j1][k] - r[j2][k];
      dr[k] = dr[k] - SignR(RegionH[k], dr[k]-RegionH[k])
        - SignR(RegionH[k], dr[k]+RegionH[k]);
      rr = rr + dr[k]*dr[k];
    }
    if (rr < rrCut) {
      ri2 = 1.0/rr; ri6 = ri2*ri2*ri2; r1 = sqrt(rr);
      fcVal = 48.0*ri2*ri6*(ri6-0.5) + Duc/r1;
      for (k=0; k<3; k++) {
        f = fcVal*dr[k];
        ra[j1][k] = ra[j1][k] + f;
        ra[j2][k] = ra[j2][k] - f;
      }
    }
  }
}
    
```

$$\begin{aligned} \text{Iteration_count} &= 1 + 2 + \dots + (N-1) \\ &= \frac{(N-1)(1+N-1)}{2} \\ &= \frac{(N-1)N}{2} = O(N^2) \end{aligned}$$

Time Discretization

Snapshots with time interval Δ : double DeltaT

$$(\vec{r}_i(0), \vec{v}_i(0)) \mapsto (\vec{r}_i(\Delta), \vec{v}_i(\Delta)) \mapsto (\vec{r}_i(2\Delta), \vec{v}_i(2\Delta)) \mapsto \dots$$



Question: How to predict the next state, $(\vec{r}_i(t + \Delta), \vec{v}_i(t + \Delta))$, from the current state, $(\vec{r}_i(t), \vec{v}_i(t))$?

Solution: Taylor expansion

$$f(x_0 + h) = \sum_{n=0}^{\infty} \frac{h^n}{n!} \frac{d^n f}{dx^n} = f(x_0) + h \left. \frac{df}{dx} \right|_{x=x_0} + \frac{h^2}{2} \left. \frac{d^2 f}{dx^2} \right|_{x=x_0} + \frac{h^3}{3!} \left. \frac{d^3 f}{dx^3} \right|_{x=x_0} + \dots$$

See the [Taylor expansion](#) note for a proof

Verlet Discretization

Position:

$$\begin{aligned} \vec{r}_i(t + \Delta) &= \vec{r}_i(t) + \vec{v}_i(t)\Delta + \frac{1}{2}\vec{a}_i(t)\Delta^2 + \frac{1}{6}\vec{\ddot{r}}_i(t)\Delta^3 + O(\Delta^4) \\ + \vec{r}_i(t - \Delta) &= \vec{r}_i(t) - \vec{v}_i(t)\Delta + \frac{1}{2}\vec{a}_i(t)\Delta^2 - \frac{1}{6}\vec{\ddot{r}}_i(t)\Delta^3 + O(\Delta^4) \end{aligned}$$

$$\vec{r}_i(t + \Delta) + \vec{r}_i(t - \Delta) = 2\vec{r}_i(t) + \vec{a}_i(t)\Delta^2 + O(\Delta^4)$$

$$\therefore \vec{r}_i(t + \Delta) = 2\vec{r}_i(t) - \vec{r}_i(t - \Delta) + \vec{a}_i(t)\Delta^2 + O(\Delta^4)$$

Velocity:

$$\begin{aligned} \vec{r}_i(t + \Delta) &= \vec{r}_i(t) + \vec{v}_i(t)\Delta + \frac{1}{2}\vec{a}_i(t)\Delta^2 + \frac{1}{6}\vec{\ddot{r}}_i(t)\Delta^3 + O(\Delta^4) \\ - \vec{r}_i(t - \Delta) &= \vec{r}_i(t) - \vec{v}_i(t)\Delta + \frac{1}{2}\vec{a}_i(t)\Delta^2 - \frac{1}{6}\vec{\ddot{r}}_i(t)\Delta^3 + O(\Delta^4) \end{aligned}$$

$$\vec{r}_i(t + \Delta) - \vec{r}_i(t - \Delta) = 2\vec{v}_i(t)\Delta + O(\Delta^3)$$

$$\therefore \vec{v}_i(t) = \frac{\vec{r}_i(t + \Delta) - \vec{r}_i(t - \Delta)}{2\Delta} + O(\Delta^2)$$

Verlet Algorithm

Verlet discretization:

$$\begin{cases} \vec{r}_i(t + \Delta) = 2\vec{r}_i(t) - \vec{r}_i(t - \Delta) + \vec{a}_i(t)\Delta^2 + O(\Delta^4) \\ \vec{v}_i(t) = \frac{\vec{r}_i(t + \Delta) - \vec{r}_i(t - \Delta)}{2\Delta} + O(\Delta^2) \end{cases}$$

Loup Verlet



Verlet algorithm:

Given $\vec{r}_i(t - \Delta)$ & $\vec{r}_i(t)$,

1. Compute $\vec{a}_i(t)$ as a function of $\{\vec{r}_i(t)\}$
2. $\vec{r}_i(t + \Delta) \leftarrow 2\vec{r}_i(t) - \vec{r}_i(t - \Delta) + \vec{a}_i(t)\Delta^2$
3. $\vec{v}_i(t) \leftarrow [\vec{r}_i(t + \Delta) - \vec{r}_i(t - \Delta)]/2\Delta$

Drawback: Positions & velocities are not simultaneously updated for the same time step

ComputeAccel(): **in** $r[i][k]$ → **out** $ra[i][k]$

```
for (n=0; n<nAtom; n++)
  for (k=0; k<3; k++) ra[n][k] = 0.0;
for (j1=0; j1<nAtom-1; j1++) {
  for (j2=j1+1; j2<nAtom; j2++) {
    for (rr=0.0, k=0; k<3; k++) {
      dr[k] = r[j1][k] - r[j2][k];
      dr[k] = dr[k] - SignR(RegionH[k], dr[k]-RegionH[k])
                - SignR(RegionH[k], dr[k]+RegionH[k]);
      rr = rr + dr[k]*dr[k];
    }
    if (rr < rrCut) {
      ri2 = 1.0/rr; ri6 = ri2*ri2*ri2; r1 = sqrt(rr);
      fcVal = 48.0*ri2*ri6*(ri6-0.5) + Duc/r1;
      for (k=0; k<3; k++) {
        f = fcVal*dr[k];
        ra[j1][k] = ra[j1][k] + f;
        ra[j2][k] = ra[j2][k] - f;
      }
    }
  }
}
```

Velocity Verlet Algorithm

Theorem: The following algebraic equation gives the same sequence of states, $(\vec{r}_i(n\Delta), \vec{v}_i(n\Delta))$, as that obtained by the Verlet discretization

$$\begin{cases} \vec{r}_i(t + \Delta) = \vec{r}_i(t) + \vec{v}_i(t)\Delta + \frac{1}{2}\vec{a}_i(t)\Delta^2 \\ \vec{v}_i(t + \Delta) = \vec{v}_i(t) + \frac{\vec{a}_i(t) + \vec{a}_i(t + \Delta)}{2}\Delta \end{cases}$$

Velocity Verlet algorithm:

Given $(\vec{r}_i(t), \vec{v}_i(t))$,

1. Compute $\vec{a}_i(t)$ as a function of $\{\vec{r}_i(t)\}$
2. $\vec{v}_i\left(t + \frac{\Delta}{2}\right) \leftarrow \vec{v}_i(t) + \frac{\Delta}{2}\vec{a}_i(t)$
3. $\vec{r}_i(t + \Delta) \leftarrow \vec{r}_i(t) + \vec{v}_i\left(t + \frac{\Delta}{2}\right)\Delta$
4. Compute $\vec{a}_i(t + \Delta)$ as a function of $\{\vec{r}_i(t + \Delta)\}$
5. $\vec{v}_i(t + \Delta) \leftarrow \vec{v}_i\left(t + \frac{\Delta}{2}\right) + \frac{\Delta}{2}\vec{a}_i(t + \Delta)$

```
void SingleStep() {
    HalfKick();
    for (int n=0; n<nAtom; n++)
        for (int k=0; k<3; k++)
            r[n][k] = r[n][k] + DeltaT*rv[n][k];
    ApplyBoundaryCond();
    ComputeAccel();
    HalfKick();
}

void HalfKick() {
    for (int n=0; n<nAtom; n++)
        for (int k=0; k<3; k++)
            rv[n][k] = rv[n][k] + DeltaTH*ra[n][k];
}
```

Velocity Verlet Algorithm for StepLimit Steps

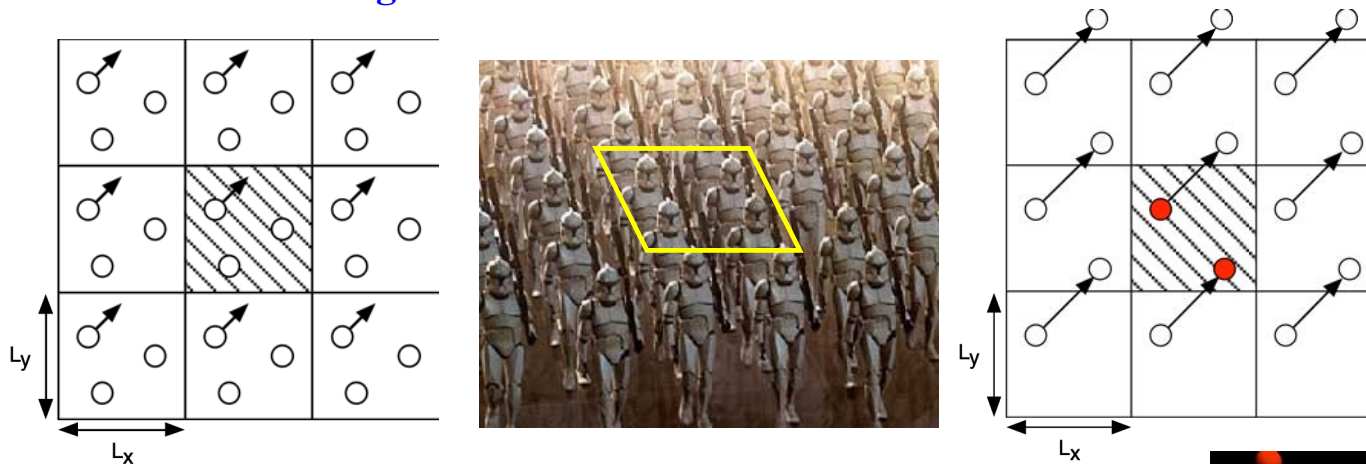
```
Initialize ( $\vec{r}_i, \vec{v}_i$ ) for all  $i$   
Compute  $\vec{a}_i$  for all  $i$  as a function of  $\{\vec{r}_i\}$  function ComputeAccel ()  
for stepCount = 1 to StepLimit  
    do the following function SingleStep ()  
         $\vec{v}_i \leftarrow \vec{v}_i + \vec{a}_i \Delta / 2$  for all  $i$   
         $\vec{r}_i \leftarrow \vec{r}_i + \vec{v}_i \Delta$  for all  $i$   
        Compute  $\vec{a}_i$  for all  $i$  as a function of  $\{\vec{r}_i\}$  function ComputeAccel ()  
         $\vec{v}_i \leftarrow \vec{v}_i + \vec{a}_i \Delta / 2$  for all  $i$   
    endfor
```

stepLimit+1 calls to
function ComputeAccel ()

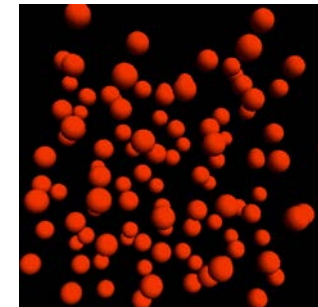
```
ComputeAccel ();  
for (stepCount=1; stepCount<=StepLimit; stepCount++) {  
    SingleStep();  
    if (stepCount%StepAvg == 0) EvalProps();  
}
```

Periodic Boundary Condition

- Allows simulation of bulk properties
- Replicate a simulation box of size $L_x \times L_y \times L_z$ to form an infinite lattice
 $\text{double Region}[3]: \text{Region}[0|1|2] = L_x|L_y|L_z$
- All images of an atom in the original simulation box move in concert



- The representative atom of each image set is defined to reside in the central simulation box
 $0 \leq x_i < L_x, 0 \leq y_i < L_y, 0 \leq z_i < L_z$
- A representative atom, which leaves the central box, returns to the central box from the other side of the box



Periodic Boundary Condition

Implementation:

$$\begin{cases} x_i \leftarrow x_i - \text{SignR}\left(\frac{L_x}{2}, x_i\right) - \text{SignR}\left(\frac{L_x}{2}, x_i - L_x\right) \\ y_i \leftarrow y_i - \text{SignR}\left(\frac{L_y}{2}, y_i\right) - \text{SignR}\left(\frac{L_y}{2}, y_i - L_y\right) \\ z_i \leftarrow z_i - \text{SignR}\left(\frac{L_z}{2}, z_i\right) - \text{SignR}\left(\frac{L_z}{2}, z_i - L_z\right) \end{cases}$$

$$\text{SignR}\left(\frac{L_x}{2}, x_i\right) = \begin{cases} \frac{L_x}{2} & x_i > 0 \\ -\frac{L_x}{2} & x_i \leq 0 \end{cases}$$

```
void ApplyBoundaryCond() {
    int n,k;
    for (n=0; n<nAtom; n++)
        for (k=0; k<3; k++)
            r[n][k]=r[n][k]
                -SignR(RegionH[k],r[n][k])
                -SignR(RegionH[k],r[n][k]-Region[k]);
}
```

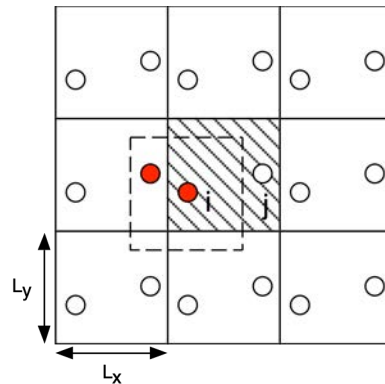
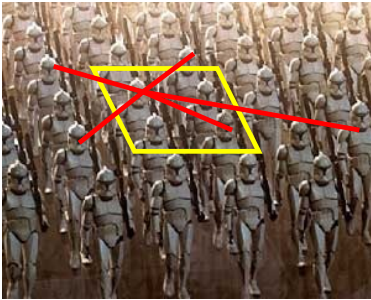
```
double SignR(double v,double x) {
    if (x > 0)
        return v;
    else
        return -v;
}
```

double RegionH[3]: An array ($L_x/2, L_y/2, L_z/2$)

Minimum Image Convention

- Each atom, i , interacts with only the image with the minimum distance among all the images of another atom, j
- Exact, if the interatomic potential has a finite range, $r_c \leq \min(L_x, L_y, L_z)/2$

$$\begin{cases} -\frac{L_x}{2} \leq x_{ij} < \frac{L_x}{2} \\ -\frac{L_y}{2} \leq y_{ij} < \frac{L_y}{2} \\ -\frac{L_z}{2} \leq z_{ij} < \frac{L_z}{2} \end{cases} \quad \begin{cases} x_{ij} \leftarrow x_{ij} - \text{SignR}\left(\frac{L_x}{2}, x_{ij} + \frac{L_x}{2}\right) - \text{SignR}\left(\frac{L_x}{2}, x_{ij} - \frac{L_x}{2}\right) \\ y_{ij} \leftarrow y_{ij} - \text{SignR}\left(\frac{L_y}{2}, y_{ij} + \frac{L_y}{2}\right) - \text{SignR}\left(\frac{L_y}{2}, y_{ij} - \frac{L_y}{2}\right) \\ z_{ij} \leftarrow z_{ij} - \text{SignR}\left(\frac{L_z}{2}, z_{ij} + \frac{L_z}{2}\right) - \text{SignR}\left(\frac{L_z}{2}, z_{ij} - \frac{L_z}{2}\right) \end{cases}$$

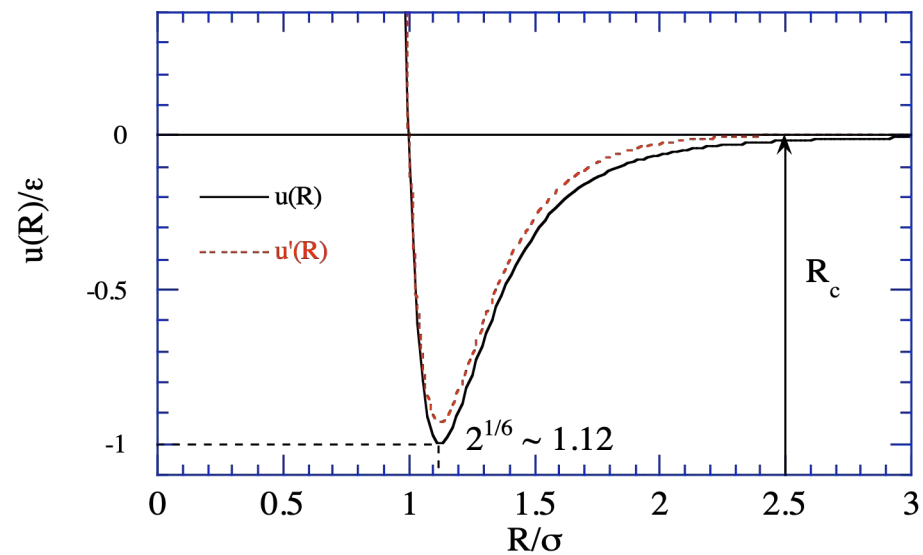
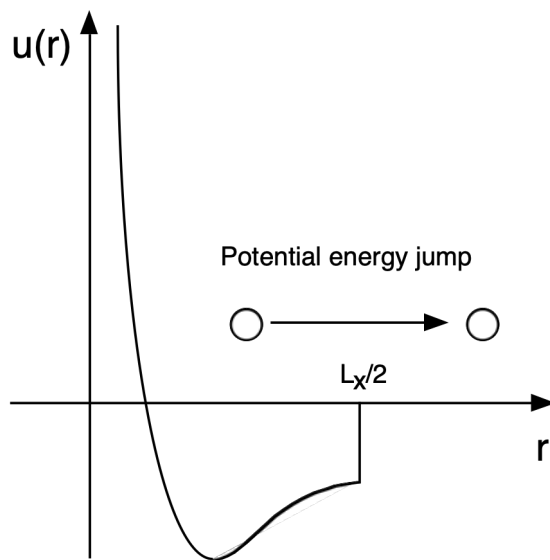


```
for (j1=0; j1<nAtom-1; j1++) {
  for (j2=j1+1; j2<nAtom; j2++) {
    for (rr=0.0, k=0; k<3; k++) {
      dr[k] = r[j1][k] - r[j2][k];
      dr[k] = dr[k] - SignR(RegionH[k], dr[k]-RegionH[k])
                - SignR(RegionH[k], dr[k]+RegionH[k]);
      rr = rr + dr[k]*dr[k];
    }
    if (rr < rrCut) {
      ri2 = 1.0/rr; ri6 = ri2*ri2*ri2; r1 = sqrt(rr);
      fcVal = 48.0*ri2*ri6*(ri6-0.5) + Duc/r1;
      for (k=0; k<3; k++) {
        f = fcVal*dr[k];
        ra[j1][k] = ra[j1][k] + f;
        ra[j2][k] = ra[j2][k] - f;
      }
    }
  }
}
```

Shifted Potential

- Make the minimum image convention exact
- Truncate $u(r)$ such that both it & its derivative du/dr are continuous at $r = r_c$

$$u'(r) = \begin{cases} u(r) - u(r_c) - (r - r_c) \left. \frac{du}{dr} \right|_{r=r_c} & r < r_c \\ 0 & r \geq r_c \end{cases}$$



#define RCUT 2.5

Force Calculation in md.c

```

for (n=0; n<nAtom; n++)
  for (k=0; k<3; k++) ra[n][k] = 0.0;
for (j1=0; j1<nAtom-1; j1++) {
  for (j2=j1+1; j2<nAtom; j2++) {
    for (rr=0.0, k=0; k<3; k++) {
      dr[k] = r[j1][k] - r[j2][k];
      dr[k] = dr[k] - SignR(RegionH[k], dr[k]-RegionH[k])
                - SignR(RegionH[k], dr[k]+RegionH[k]);
      rr = rr + dr[k]*dr[k];
    }
    if (rr < rrCut) {
      ri2 = 1.0/rr; ri6 = ri2*ri2*ri2; r1 = sqrt(rr);
      fcVal = 48.0*ri2*ri6*(ri6-0.5) + Duc/r1;
      for (k=0; k<3; k++) {
        f = fcVal*dr[k];
        ra[j1][k] = ra[j1][k] + f;
        ra[j2][k] = ra[j2][k] - f;
      }
    }
    rr = RCUT*RCUT; ri2 = 1.0/rr; ri6 = ri2*ri2*ri2; r1 = sqrt(rr);
    Uc = 4.0*ri6*(ri6-1.0);
    Duc = -48.0*ri6*(ri6-0.5)/r1;
  }
}

```

minimum image convention

shifted potential

$$\vec{a}_k(t) = \sum_{i < j} \vec{r}_{ij}(t) \left(\frac{48}{r^2} \left(\frac{1}{r^{12}} - \frac{1}{2r^6} \right) \right)_{r=|\vec{r}_{ij}(t)|} (\delta_{ik} - \delta_{jk})$$

Energy Conservation

- Total energy = kinetic energy + potential energy

$$E = K + V = \sum_{i=0}^{N-1} \frac{m}{2} |\dot{\vec{r}}_i|^2 + \sum_{i < j} u(r_{ij})$$

- Total energy is constant as a function of time, if no discretization error

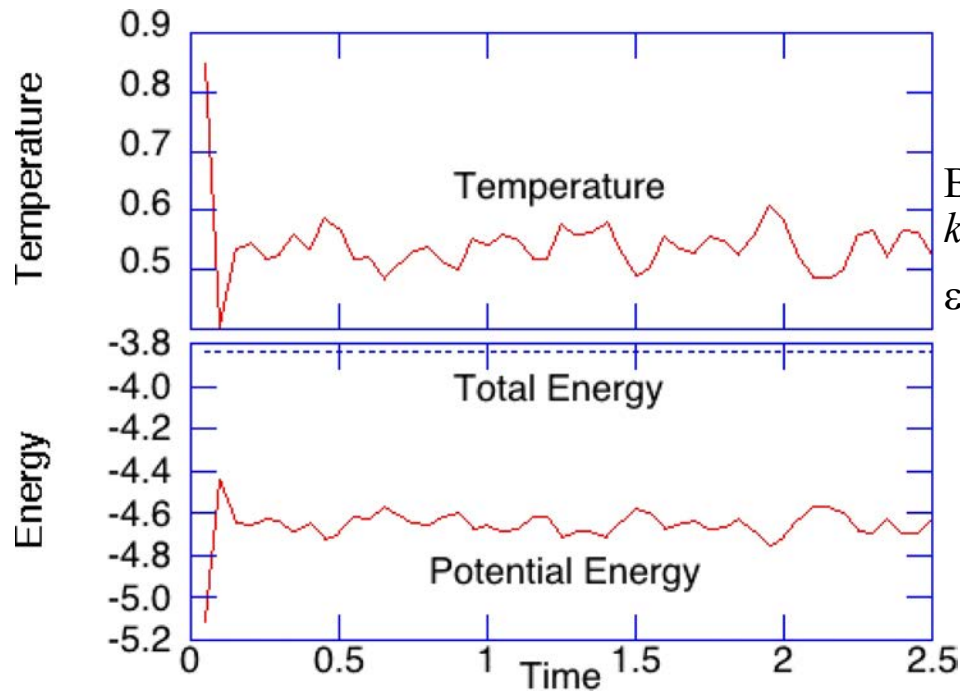
$$\begin{aligned} \dot{E} &= \dot{K} + \dot{V} \\ &= \sum_{i=0}^{N-1} \frac{m}{2} \frac{d}{dt} (\dot{x}_i^2 + \dot{y}_i^2 + \dot{z}_i^2) + \sum_{i=0}^{N-1} \left(\frac{dx_i}{dt} \frac{\partial V}{\partial x_i} + \frac{dy_i}{dt} \frac{\partial V}{\partial y_i} + \frac{dz_i}{dt} \frac{\partial V}{\partial z_i} \right) \\ &= \sum_{i=0}^{N-1} \frac{m}{2} 2(\dot{x}_i \ddot{x}_i + \dot{y}_i \ddot{y}_i + \dot{z}_i \ddot{z}_i) + \sum_{i=0}^{N-1} \left(\dot{x}_i \frac{\partial V}{\partial x_i} + \dot{y}_i \frac{\partial V}{\partial y_i} + \dot{z}_i \frac{\partial V}{\partial z_i} \right) \\ &= \sum_{i=0}^{N-1} m \dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i + \sum_{i=0}^{N-1} \dot{\vec{r}}_i \cdot \frac{\partial V}{\partial \vec{r}_i} = \sum_{i=0}^{N-1} \dot{\vec{r}}_i \cdot \left(m \ddot{\vec{r}}_i + \frac{\partial V}{\partial \vec{r}_i} \right) = 0 \end{aligned}$$

$$\vec{a} \cdot \vec{b} = (a_x, a_y, a_z) \cdot (b_x, b_y, b_z) = a_x b_x + a_y b_y + a_z b_z \text{ and } \frac{d}{dt}(fg) = \dot{f}g + f\dot{g}$$

```
double kinEnergy, potEnergy, totEnergy;
```

Energy Conservation

md.c: $\Delta = 0.005$



Temperature, T

$$\frac{3Nk_B T}{2} = \sum_{i=0}^{N-1} \frac{m}{2} |\dot{\vec{r}}_i|^2$$

Boltzmann constant:
 $k_B = 1.38062 \times 10^{-16}$ erg/Kelvin
 $\varepsilon/k_B = 120$ Kelvin

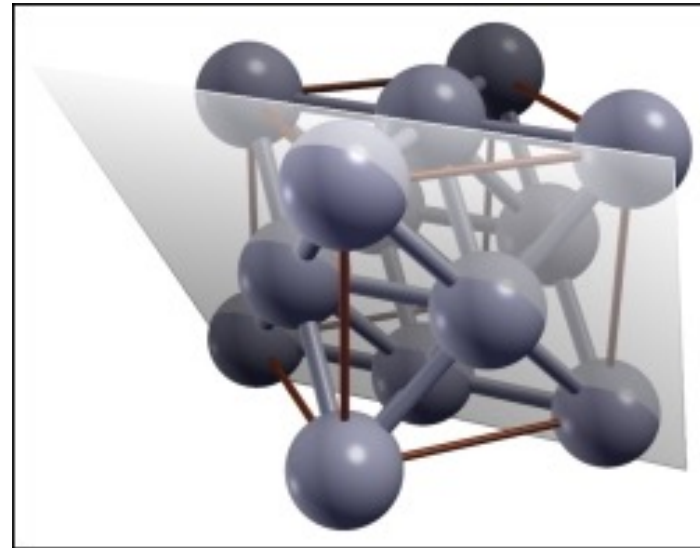
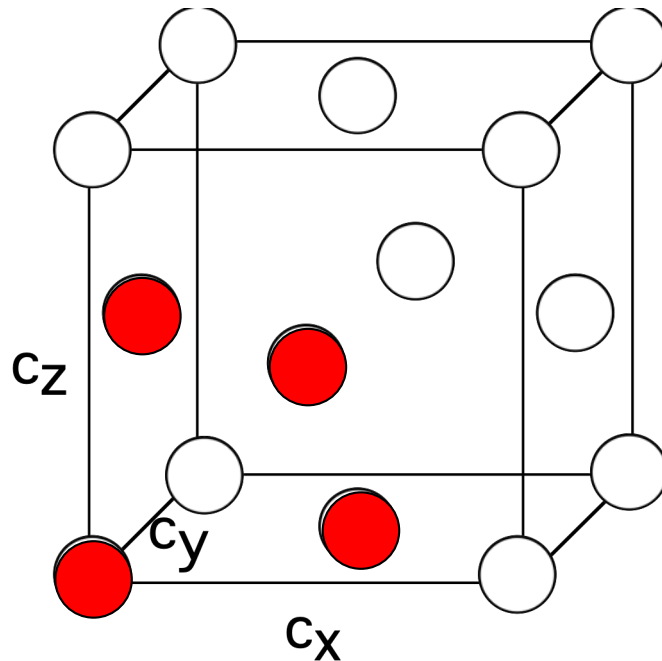
Normalized form

$$\frac{T}{\varepsilon/k_B} = \frac{1}{3N} \sum_{i=0}^{N-1} |\dot{\vec{r}}_i'|^2$$

double temperature;

Initial Configuration

- Face centered cubic (FCC) lattice: 4 atoms per unit cell, c^3
- Density $\rho = 4/c^3$



double gap[3]: (c_x, c_y, c_z), the edge lengths of the unit cell
double Density: Input parameter, input
int InitUcell[3]: # of unit cells in the x, y , and z directions, input
nAtom = 4 × InitUcell[0] × InitUcell[1] × InitUcell[2]

Initial Velocities

- Generate random velocities of magnitude, v_0

$$\frac{v_0^2}{3} = T_{init} \Rightarrow v_0 = \sqrt{3T_{init}}$$

- Random velocity

$$\vec{v}_i = v_0(\xi_0, \xi_1, \xi_2) = v_0 \vec{\xi}$$

where the randomly oriented vector of unit length is $\vec{\xi}$

- Algorithm

Let rand() be a uniform random-number generator in the range [0, 1]

1. $\zeta_i = 2 \times \text{rand}() - 1$ ($i = 0, 1$) so that $-1 \leq \zeta_i < 1$
2. $s^2 = \zeta_0^2 + \zeta_1^2$
3. if $s^2 < 1$, accept $(\xi_0, \xi_1, \xi_2) = (2\sqrt{1-s^2}\zeta_0, 2\sqrt{1-s^2}\zeta_1, 1-2s^2)$
4. else reject and go to 1

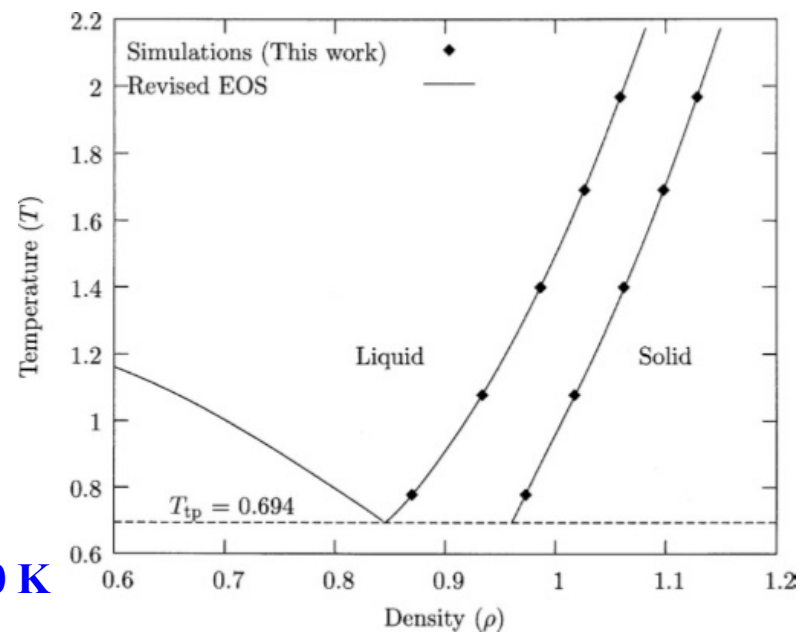
double InitTemp: Initial temperature, input

MD Input Parameters

md.in

3 3 3 **InitUcell[3]**
 0.8 **Density**
 1.0 **InitTemp**
 0.005 **DeltaT**
 10000 **StepLimit**
 10 **StepAvg**

in 0.0254 Å³ for Ar
 in 120 K
 in 2.2 ps



Compare runtime of md.c for InitUcell[] = {3,3,3} vs. {10,10,10}
 108 atoms 4000 atoms

Recap: MD Computation

$\{\vec{a}_i | i = 0, \dots, N-1\} \leftarrow \text{ComputeAccel}(\{\vec{r}_i\})$

Repeat *StepLimit* times

1. $\vec{v}_i \leftarrow \vec{v}_i + \frac{\Delta}{2} \vec{a}_i$
2. $\vec{r}_i \leftarrow \vec{r}_i + \vec{v}_i \Delta$
3. $\{\vec{a}_i\} \leftarrow \text{ComputeAccel}(\{\vec{r}_i\})$
4. $\vec{v}_i \leftarrow \vec{v}_i + \frac{\Delta}{2} \vec{a}_i$

DeltaT

ComputeAccel(): r[][] → ra[][]

```
for (i=0; i<N; i++)  
  for (k=0; k<3; k++) ra[i][k] = 0.0;  
  for (i=0; i<N-1; i++) {  
    for (j=i+1; j<N; j++) {  
      ...  
      for (k=0; k<3; k++) {  
        ...  
        ra[i][k] = ra[i][k] + fk;  
        ra[j][k] = ra[j][k] - fk;  
      }  
    }  
  }  
}
```

O(N²) floating-point operations

MD bottom line: Keep moving atoms ← kicked by neighbor atoms

Linked-List Cell MD

Aiichiro Nakano

*Department of Computer Science
Department of Physics & Astronomy
Department of Quantitative & Computational Biology
University of Southern California*

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Objectives:

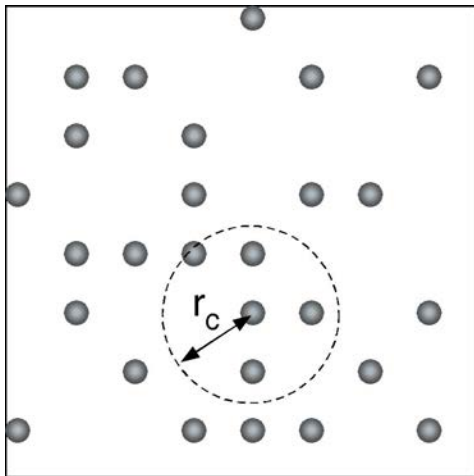
- 1. Reduced complexity, $O(N^2) \rightarrow O(N)$, for large problems**
- 2. Migration path to parallel MD — “data locality”!**

See program [lmd.c](#)



Linked-List Cell Molecular Dynamics

- Computational complexity of ComputeAccel() in md.c: $\propto N(N-1)/2 = O(N^2)$
- Data locality (cut-off length, r_c) reduces the complexity to $O(N)$: $N \times (4\pi/3)r_c^3 \times (N/V)$
- $O(N)$ algorithm uses: (1) spatially localized cells; & (2) linked lists to keep track of atoms' cell membership



```
for (j1=0; j1<nAtom-1; j1++) {  
  for (j2=j1+1; j2<nAtom; j2++) {  
    for (rr=0.0, k=0; k<3; k++) {  
      dr[k] = r[j1][k]-r[j2][k];  
      dr[k] = dr[k]-SignR(RegionH[k],dr[k]-RegionH[k])  
              -SignR(RegionH[k],dr[k]+RegionH[k]);  
      rr = rr+dr[k]*dr[k];  
    }  
    if (rr < rrCut) {  
      rrcut = r_c  
      ri2 = 1.0/rr; ri6 = ri2*ri2*ri2; r1 = sqrt(rr);  
      fcVal = 48.0*ri2*ri6*(ri6-0.5)+Duc/r1;  
      for (k=0; k<3; k++) {  
        f = fcVal*dr[k];  
        ra[j1][k] = ra[j1][k]+f;  
        ra[j2][k] = ra[j2][k]-f;  
      }  
    }  
  }  
}
```

DIY: Compare the runtime of md.c & lmd.c!

Cell Data Structures

- Cell size**

$$L_{c\alpha} = \lfloor L_{\alpha} / r_c \rfloor \quad (\alpha = x, y, z) \quad (\text{int } lc[3])$$

$$r_{c\alpha} = L_{\alpha} / L_{c\alpha} \quad (\text{double } rc[3])$$

where

L_{α} : simulation box length
(double Region[3])

r_c : Cut-off length (RCUT)

- Vector cell index, $0 \leq c_{\alpha} \leq L_{c\alpha} - 1$**

- Serial cell index:**

$$c = c_x L_{cy} L_{cz} + c_y L_{cz} + c_z$$

or

$$c_x = \lfloor c / (L_{cy} L_{cz}) \rfloor$$

$$c_y = \lfloor c / L_{cz} \rfloor \bmod L_{cy}$$

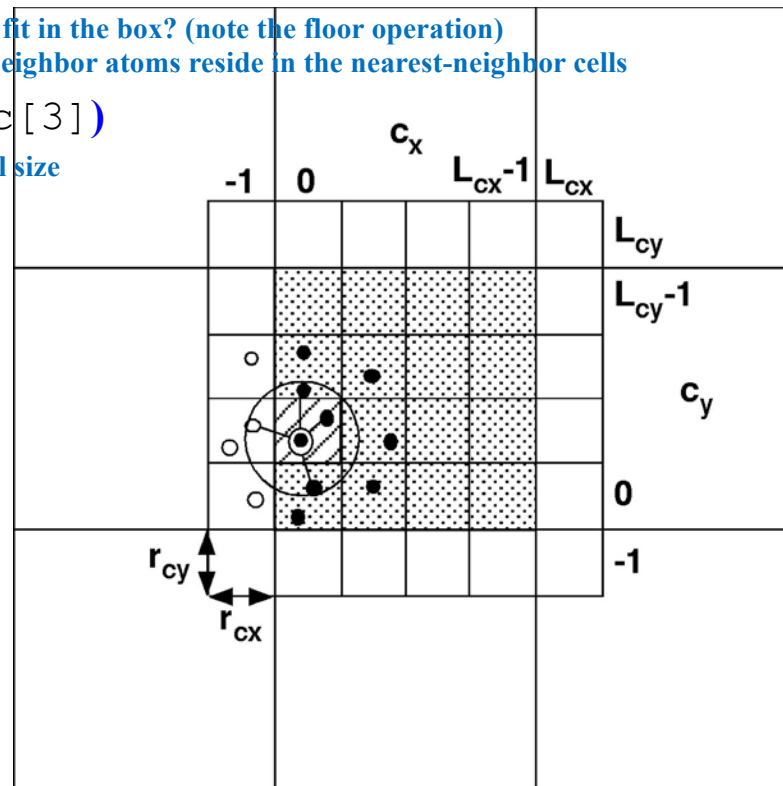
$$c_z = c \bmod L_{cz}$$

- Atom-to-cell mapping:**

$$c_{\alpha} = \lfloor r_{\alpha} / r_{c\alpha} \rfloor$$

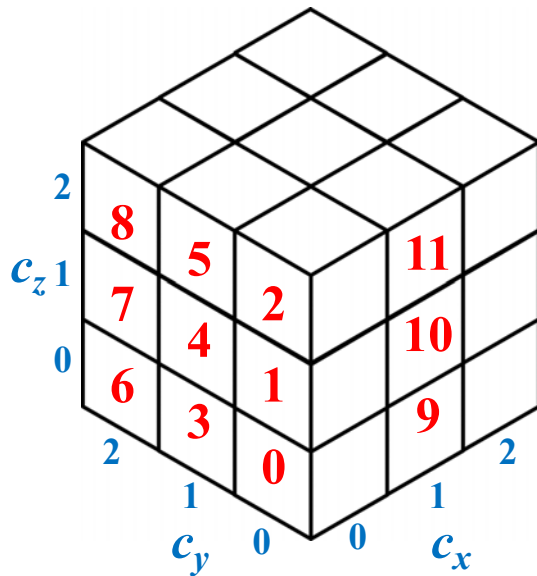
How many cutoff lengths fit in the box? (note the floor operation)

This guarantees that all neighbor atoms reside in the nearest-neighbor cells



$$2 \times \text{\$1} + 3 \times \text{\$0.01} + 4 \times \text{\$0.01} = 234\text{\cent}$$

Cell Index Example



How many cells are before me?
 $c = c_x L_{cy} L_{cz} + c_y L_{cz} + c_z$

or

$$\begin{aligned} c_x &= \lfloor c / (L_{cy} L_{cz}) \rfloor \\ c_y &= \lfloor c / L_{cz} \rfloor \bmod L_{cy} \\ c_z &= c \bmod L_{cz} \end{aligned}$$

Which column, row, & height?

$$L_{cx} = L_{cy} = L_{cz} = 3$$

c	cx	cy	cz
0	0	0	0
1	0	0	1
2	0	0	2
3	0	1	0
4	0	1	1
5	0	1	2
6	0	2	0
7	0	2	1
8	0	2	2
9	1	0	0
10	1	0	1
11	1	0	2

reset

...

slow ← fast

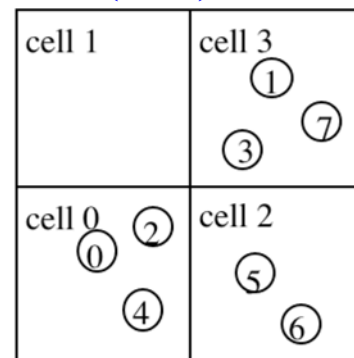


Linked Lists

Data Structures:

`lsc1[NMAX]` : **Linked lists**; `lsc1[i]` holds the atom index to which the *i*-th atom points.

`head[NCLMAX]` : **head[c]** holds the index of the first atom in the *c*-th cell, or **head[c] = EMPTY (= -1)** if there is no atom in the cell.



	0	1	2	3				
head	4	E	6	7				

	0	1	2	3	4	5	6	7
lsc1	E	E	0	1	2	E	5	3

For details, see the lecture note on
“linked-list cell MD algorithm”

<https://aiichironakano.github.io/cs596/01-1LinkedListCell.pdf>

Example of cell 3



Linked List Construction Algorithm

```

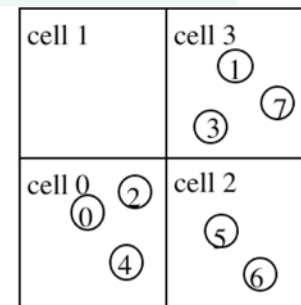
/* Reset the headers, head */
for (c=0; c<lcxyz; c++) head[c] = EMPTY;
/* Scan atoms to construct headers, head, & linked lists, lscl */
for (i=0; i<nAtom; i++) {
    /* Vector cell index to which this atom belongs */
    for (a=0; a<3; a++) mc[a] = r[i][a]/rc[a];
    /* Translate the vector cell index, mc, to a scalar cell index */
    c = mc[0]*lcxyz+mc[1]*lc[2]+mc[2];
    /* Link to the previous occupant (or EMPTY if you're the 1st) */
    lscl[i] = head[c];
    head[c] = i;
}

```

where
 $lcxyz = lc[1]*lc[2]$
 $lcxyz = lcxyz*lc[0]$



...



	0	1	2	3	4	5	6	7
head	4	E	6	7				
lscl	E	E	0	1	2	E	5	3



$O(N)$ Force Calculation Algorithm

```
for (mc[0]=0; mc[0]<lc[0]; (mc[0])++)  
for (mc[1]=0; mc[1]<lc[1]; (mc[1])++)  $O(N)$   
for (mc[2]=0; mc[2]<lc[2]; (mc[2])++) { /* Scan all cells */  
    c = mc[0]*lcyz+mc[1]*lc[2]+mc[2]; /* Calculate a scalar cell index */  
    for (mc1[0]=mc[0]-1; mc1[0]<=mc[0]+1; (mc1[0])++)  $3^3 = 27$   
    for (mc1[1]=mc[1]-1; mc1[1]<=mc[1]+1; (mc1[1])++)  
    for (mc1[2]=mc[2]-1; mc1[2]<=mc[2]+1; (mc1[2])++) { /* Scan neighbor cells */  
        for (a=0; a<3; a++) { /* Unwrapping the periodic boundary condition */  
            if (mc1[a] < 0)  
                rshift[a] = -Region[a];  
            else if (mc1[a]>=lc[a])  
                rshift[a] = Region[a]; /* Taking care of periodic boundary condition */  
            else  
                rshift[a] = 0.0;  
        }  
        c1 = ((mc1[0]+lc[0])%lc[0])*lcyz  
            + ((mc1[1]+lc[1])%lc[1])*lc[2]  
            + ((mc1[2]+lc[2])%lc[2]); /* Scalar cell index of the neighbor cell */  
        i = head[c]; /* Scan atom i in cell c */  
        while (i != EMPTY) {  
            j = head[c1]; /* Scan atom j in cell c1 */  
            while (j != EMPTY) {  
                if (i < j) { /* Avoid double counting of pair (i, j) */  
                     $r_{ij} = r_i - (r_j + r_{\text{shift}})$ ; /* Image-corrected relative pair position */  
                    if ( $r_{ij} < r_c$ )  
                        Compute forces on pair (i, j)  
                }  
                j = lscl[j];  
            }  
            i = lscl[i];  
        }  
    }  
}
```

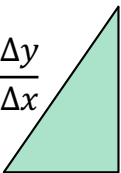
Complexity of Linked-List Cell MD

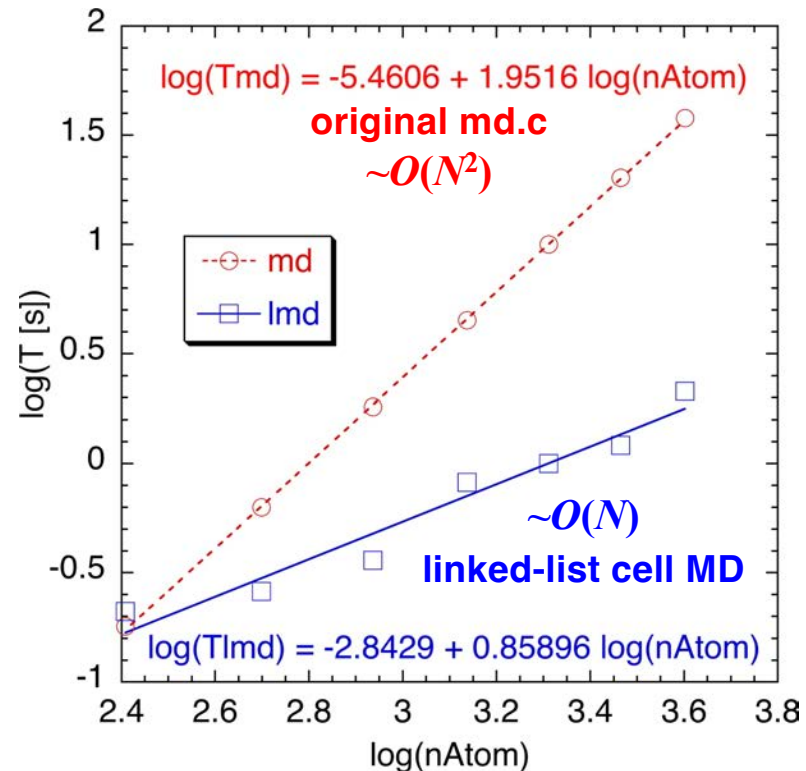
Learn computational complexity analysis by runtime measurement

See lecture note on [asymptotic analysis of functions](#)

- Power of polynomial complexity can be estimated by the slope of linear fit in **log-log plot** of runtime T vs. problem size N

$$T = cN^\alpha$$
$$\underbrace{\log T}_y = \underbrace{\log c}_\beta + \underbrace{\alpha \log N}_x$$

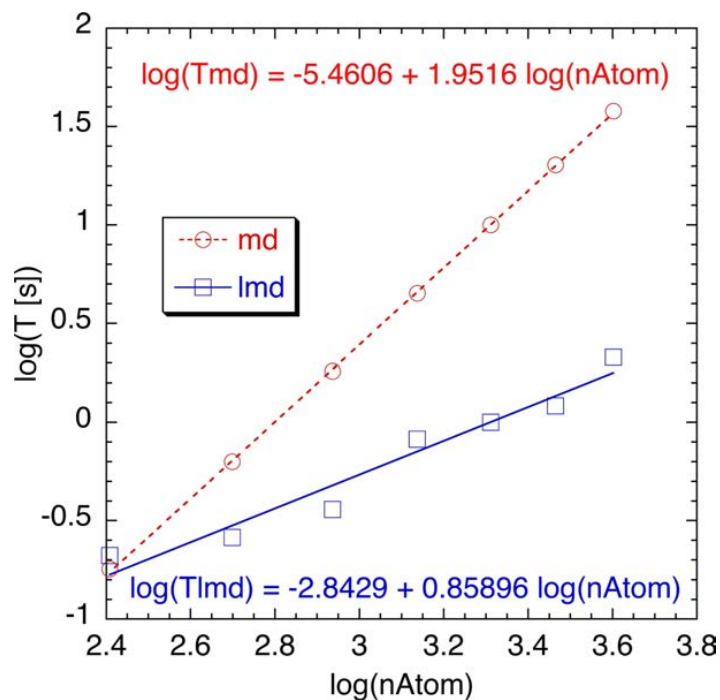
$$\alpha = \frac{\Delta y}{\Delta x}$$




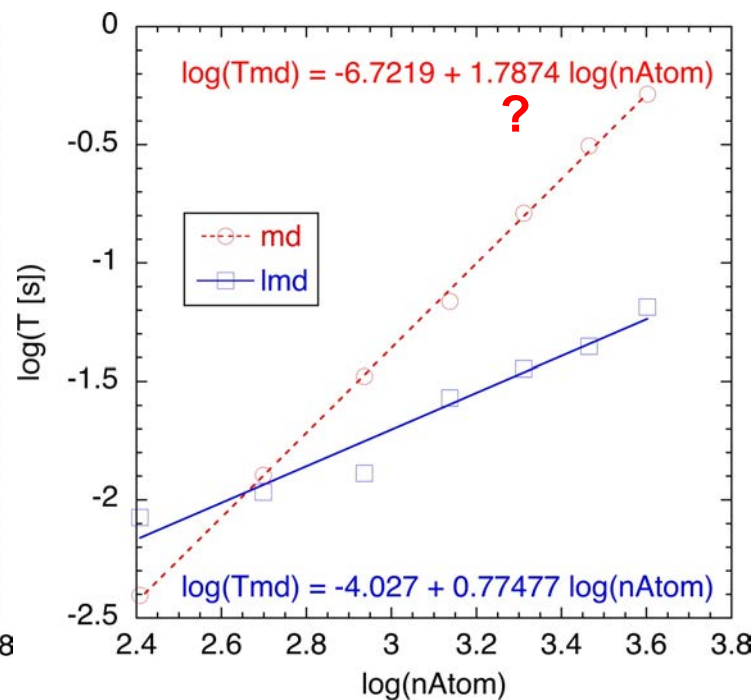
See lecture note on [Least square fit of a line](#)

Machine Dependence?

2.53 GHz Intel Core2



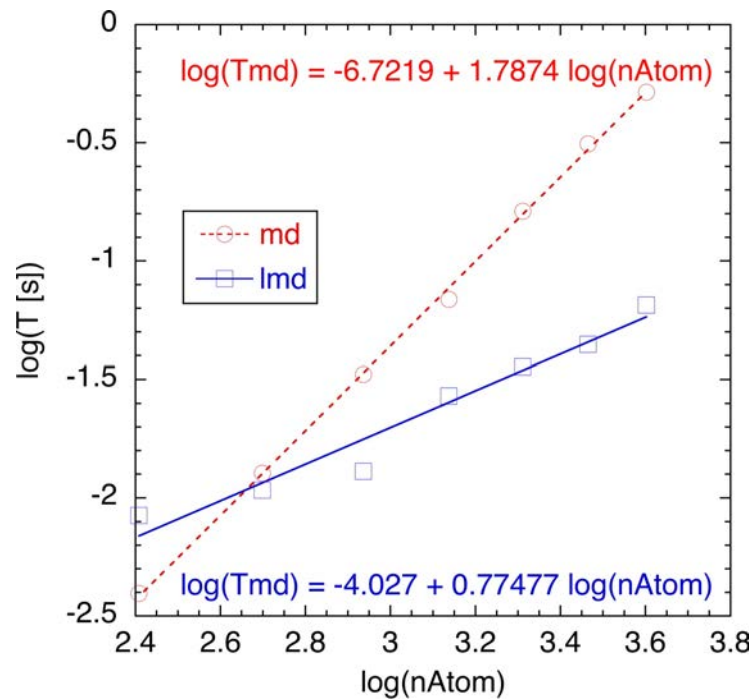
2.7 GHz Intel Core i7



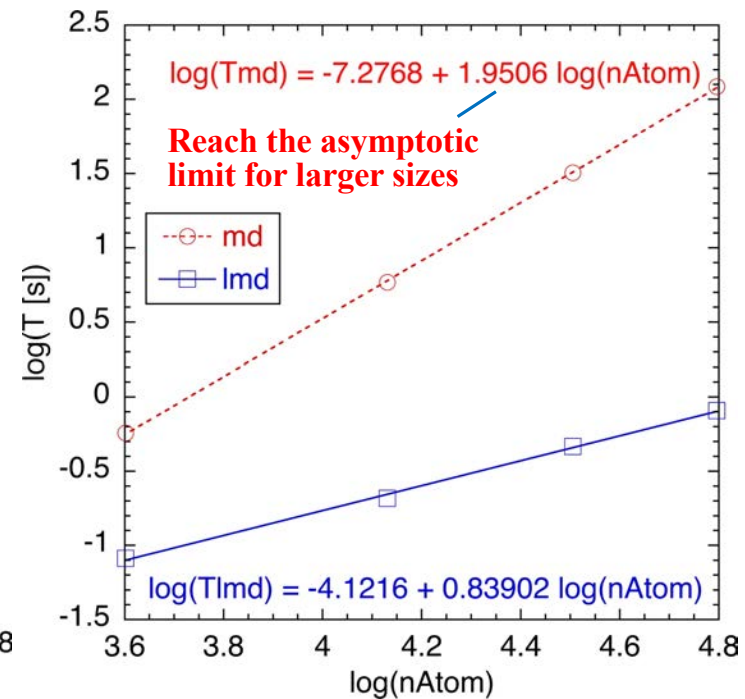
nAtom = 256, ..., 4000

Finite-Size Effect?

2.7 GHz Intel Core i7

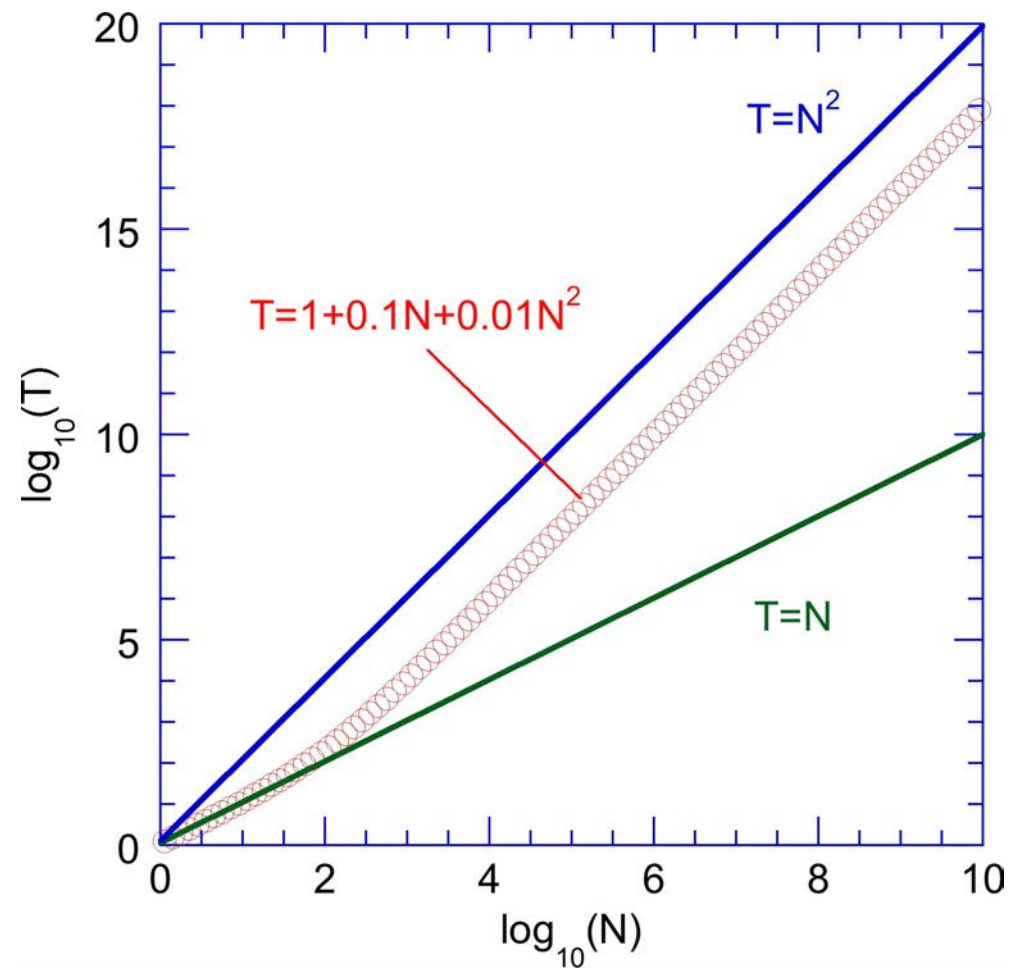


$n\text{Atom} = 256, \dots, 4000$



$n\text{Atom} = 4000, \dots, 62500$

Asymptotic Complexity?



Flop/s Performance of lmd.c

Program: lmd_sqrt_flop.c

Compile option: -O (default optimization)

```
cc -O -o lmd_sqrt_flop lmd_sqrt_flop.c -lm
```

==== Discovery cluster ====

Processor: 2.1 GHz Intel Xeon

Execution time (s) = 1.500000e-01

Number of FP operations = 1.609487e+08

MFlops rate = 1.072992e+03 (1.07 Gflop/s)

==== MacBook Pro ====

Processor: 2.3 GHz Intel Core i9

Execution time (s) = 6.417400e-02

Number of FP operations = 1.609487e+08

MFlops rate = 2.508005e+03 (2.51 Gflop/s)

Flop/s =
floating-point
operations/second

M (mega)	= 10⁶
G (giga)	= 10⁹
T (Tera)	= 10¹²
P (Peta)	= 10¹⁵
X (Exa)	= 10¹⁸

Where to Go from Here

- **Mathematical details skipped in the slides can be found in the lecture notes:**

<https://aiichironakano.github.io/cs596/01MD.pdf>

<https://aiichironakano.github.io/cs596/01-1LinkedListCell.pdf>

- **Practical application of molecular dynamics simulations in materials science is covered by the following courses:**

MASC 575: *Basics of Atomistic Simulation of Materials*

MASC 576: *Molecular Dynamics Simulations of Materials & Processes*